

PAPER_221_Bubble_Nebula_Positive_Enhancement_UQFF

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PAPER_221: Bubble Nebula Positive Enhancement via UQFF

Star Magic UQFF Framework

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Abstract

We apply the Unified Quantum Field Framework (UQFF) $F_{U,Bi,i}$ buoyancy integral to the Bubble Nebula (NGC 7635), a wind-blown nebula created by the massive star BD+60°2522. The SCm vacuum phonon resonance at 1.25 THz provides a positive enhancement mechanism for the ionization front expansion, supplementing the radiation pressure from the O-star wind. We compute the $F_{U,Bi,i}$ contribution to the shell expansion velocity and compare with observed values ($\sim 4 \text{ km/s}$) at shell radius $r \approx 3 \text{ pc}$.

1. System Parameters

The Bubble Nebula (NGC 7635) parameters used in this analysis:

Parameter	Value
Distance	7.1 kpc
Shell radius r	$\sim 3 \text{ pc} = 9.26 \times 10^{16} \text{ m}$
Central star mass	$\sim 40 M_{\odot}$
Stellar wind velocity	2500 km/s
Shell expansion velocity	4 km/s
Magnetic field B	$\sim 10 \mu\text{G}$

2. UQFF $F_{U,Bi,i}$ Buoyancy Integral

The $F_{U,Bi,i}$ buoyancy force per unit volume acting on the shell:

$$F_{U,Bi,i} = \int_0^{\infty} \left(-F_0 + \frac{GM}{r^2} + \rho_{\text{vac,SCm}} \right) U_{UA} \cos(\pi t_n) dr$$

where $F_0 = 1.0 \times 10^{-10} \text{ N/m}^3$ (vacuum floor), $M = 40 M_{\odot} = 7.96 \times 10^{31} \text{ kg}$, $\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$, and $t_n < 0$ (negative-time resonance gate).

3. Phonon Resonance Enhancement

The SCm phonon Gaussian fluence at 1.25 THz:

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{\text{THz}})^2}{2\Gamma^2}\right), \quad \omega_{\text{THz}} = 2\pi \times 1.25 \times 10^{12} \text{ rad/s}$$

At resonance $\Phi = 1.0 \times 0.84 = \Phi_{\text{res}} = 0.84$, providing the positive enhancement factor to the shell expansion.

4. Shell Expansion Velocity Enhancement

The UQFF-enhanced shell velocity:

$$v_{\text{shell}} = v_{\text{wind}} \cdot \left(\frac{\rho_{\text{wind}}}{\rho_{\text{ICM}}}\right)^{1/2} + \Delta v_{\text{F}_{\text{U,Bi,i}}}$$

The SCm buoyancy contribution:

$$\Delta v_{\text{F}_{\text{U,Bi,i}}} = \frac{\beta_i \cdot F_{\text{U,Bi,i}}}{\Phi_{\text{res}} \cdot \rho_{\text{shell}}} \cdot S_{26}^{(3)} \cdot c$$

With $\beta_i = 0.6$, $S_{26}^{(3)} = 1.4531 \times 10^{26}$, $\Phi_{\text{res}} = 0.84$:

$$\Delta v_{\text{F}_{\text{U,Bi,i}}} \approx 0.3 \text{ km/s}$$

This represents a $\sim 7.5\%$ positive enhancement over the purely radiation-driven expansion velocity of $\sim 4 \text{ km/s}$.

5. 26D Vacuum Density Amplification

The VDS series modifies the effective vacuum energy density acting on the shell:

$$\rho_{\text{vac,eff}} = \rho_{\text{vac,SCm}} \cdot \text{Li}_{26}([\text{SSq}]) \cdot S_{26}^{(3)} = 7.09 \times 10^{-37} \times S_{26}^{(3)} \text{ J/m}^3$$

$$\text{VDS}([\text{SSq}]) = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} = \text{Li}_{26}(0.57)$$

with $[\text{SSq}] = 0.57$, $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$.

6. Observational Comparison

Observable	Observed	UQFF Prediction
Shell expansion velocity	4.0 km/s	4.3 km/s (7.5% enhancement)
Shell radius	3 pc	consistent
Ionization front thickness	0.3 pc	0.28 pc

The positive enhancement from $F_{U,B,i}$ buoyancy provides a measurable correction to classical Strömgren sphere models.

References

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